

Functional Requirements Specification (FRS)

System: Jima — CARAVAN Lounge Restaurant Management System
Document: FRS v1.3 · **Updated:** August 30, 2026
Related documents: SRS (02), SDD (03), User Manual (04), Handoff (05), deployment guide (06), and operational runbooks (07-12)

1. Purpose

This document defines the functional requirements of the Jima Restaurant Management System — a full-stack web application that manages the daily operations of CARAVAN Lounge: ordering, kitchen workflow, payments, inventory, staff, branches, and reporting.

Verified implementation baseline: the registered Jima web artifact runs the real Next.js 15 / React 19 frontend in `artifacts/nextjs-app`. Browser requests use `/backend/api/*`, which Next.js rewrites to the NestJS service's `/api/*`. NestJS and TypeORM own all PostgreSQL access. The separate `artifacts/api-server` is not the restaurant application API.

Requirements versus evidence: the items below define required behavior. A requirement must not be treated as production-verified unless it appears as passed evidence in document 12.

2. User Roles

Role	Description
Admin	Full system access; manages all data, users, and configuration.

Owner	Business owner; same operational visibility as manager, including analytics dashboard.
Manager	Runs daily operations; manages menu, staff, tables, branches, inventory, and reports.
Coordinator	Confirms incoming orders and assigns them to chefs; can cancel orders.
Waiter	Takes orders at tables (POS), adds items to open orders, tracks order progress.
Chef	Works the kitchen board: accepts, prepares, and marks orders ready.
Cashier	Collects payments, closes orders, views sales and items-purchased reports.
Branch Storekeeper	A storekeeper assigned to a branch; manages that branch's stock and requests stock from the restaurant's Main Store for the assigned branch.
Main Store Storekeeper	A storekeeper assigned to a restaurant but to no branch; receives central stock and exclusively fulfills approved Branch Stock Requests.

3. Functional Requirements

3.1 Authentication & Authorization

ID	Requirement
FR-A01	Users log in with e-mail and password; the system returns a JWT access token.
FR-A02	Every protected API call requires a valid <code>Authorization: Bearer</code> token. Login and health endpoints are intentional public exceptions; administrative seed operations must remain protected.
FR-A03	Menu entries and pages are restricted per role (role-based access control).

FR-A04	Passwords are stored hashed (bcrypt), never in plain text.
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3.2 Orders & POS

ID	Requirement
FR-O01	Waiters create orders through a POS screen: pick a table (or take-away), select menu items by category, set quantities and notes.
FR-O02	Orders follow the lifecycle: pending → confirmed → preparing → ready → served → paid , with cancelled possible before "ready".
FR-O03	The coordinator confirms a pending order and assigns a chef to it.
FR-O04	Waiters may append items to any order that is not yet paid or cancelled.
FR-O05	The orders list shows table, chef, item count, time, total, payment method, and status; rows open a detail view.
FR-O06	Orders can be filtered by status, waiter, table, item, date, time, and payment method.
FR-O07	The list and open detail view refresh automatically every 10 seconds.
FR-O08	On phone-sized screens, the POS keeps menu and checkout controls usable by stacking the menu above the cart, preserving horizontal category browsing, and providing separate vertical scrolling for menu items and checkout content.

3.3 Kitchen

ID	Requirement
FR-K01	The kitchen board shows confirmed orders queued per chef, with tabs per chef.
FR-K02	Chefs move orders through accept → preparing → ready.
FR-K03	The Order Progress Board displays four live columns: Requested, Received & Confirmed, In Progress, Completed (ready only).

FR-K04	Orders waiting more than 20 minutes are visually flagged.
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3.4 Payments

ID	Requirement
FR-P01	Cashiers record payments against served orders with method cash, card, or wallet, including change given.
FR-P02	Paying an order marks it paid and frees its table.
FR-P03	Daily reports summarize revenue, transaction counts, and per-method breakdowns.

3.5 Menu Management

ID	Requirement
FR-M01	Managers maintain menu categories and items (name, price, description, availability, preparation time).
FR-M02	Unavailable items are hidden from the POS.

3.6 Inventory & Store

ID	Requirement
FR-I01	Inventory items track unit, current stock, minimum stock, reorder point, unit cost, category, and expiry date.
FR-I02	Low-stock alerts flag items below minimum (critical when at or below half the minimum, or zero).
FR-I03	Suppliers and purchase orders are managed with status tracking.
FR-I04	Stock adjustments record additions, deductions, and waste with reasons and the acting user.
FR-I05	The existing generic Item Requests workflow remains available: staff submit requests against branch inventory; authorized users approve, reject, issue, and track them, with low-stock / expiry

	badges shown on request rows. It is separate from, and has not been replaced by, Branch Stock Requests.
FR-I06	Only a Branch Storekeeper (a storekeeper with a branch assignment) may create a Branch Stock Request. Its destination is always that user's assigned branch; request lines select Main Store items and positive quantities.
FR-I07	A Manager assigned to the destination branch or an Owner of the same restaurant may approve or reject a pending Branch Stock Request. A request or decision records workflow state and audit data but never changes Main Store or branch stock.
FR-I08	Only the branchless Main Store Storekeeper of the same restaurant may fulfill an approved request. Fulfillment must reject non-approved requests, missing destinations or items, and insufficient Main Store stock.
FR-I09	Fulfillment is one atomic database transaction: for every line it decrements the restaurant's Main Store item and increments the matching name-and-unit inventory item in the assigned destination branch, creating that branch item when needed. Any failure rolls back the complete transfer.
FR-I10	The Branch Stock Request audit trail records requester and request time; approver or rejector and decision time; fulfilling Main Store Storekeeper and fulfillment time; a linked Main Store stock-out movement; a linked destination-branch addition adjustment; and the Main Store and branch balances after each transferred line.
FR-I11	Main Store items, requests, decisions, fulfillment, movements, and destination inventory are isolated by restaurant. Branch Storekeepers and Managers see only requests for their assigned branch; Managers cannot decide another branch's request, and no transfer may target a branch outside the restaurant.

3.6.1 Branch Stock Request lifecycle

Status / action	Authorized actor	Stock effect
pending — request submitted	Destination Branch Storekeeper	None

approved or rejected	Destination Manager or restaurant Owner	None
approved → transferred	Branchless Main Store Storekeeper only	Atomic Main Store stock-out and destination-branch stock-in

3.7 Staff, Branches & Tables

ID	Requirement
FR-S01	Managers create and manage staff accounts with roles and branch assignment.
FR-S02	Restaurants and branches are managed hierarchically (restaurant → branches).
FR-S03	Tables have number, capacity, section, and live status (available / occupied).

3.8 Dashboards & Reports

ID	Requirement
FR-D01	Admin/Owner/Manager see a combined dashboard: today's KPIs (sales, orders, average order value, in-progress, low stock, stock value), sales trend, top sellers, profitability, and kitchen performance — on a single page. KPI cards open detail popups.
FR-D02	Reports offer per-area export tabs (Order Board, Menu, Inventory, Item Requested, Staff, Branches) with filters and Excel / PDF export.
FR-D03	Cashiers see a reduced report view: Sales and Items Purchased only, filterable by type and date, exportable to Excel / PDF.
FR-D04	Role-specific landing: waiters, coordinators, and cashiers land on Orders/POS instead of the dashboard.

3.9 Notifications

ID	Requirement
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FR-N01

The system generates notifications targeted at roles or specific users (e.g. new order, order ready) with read tracking.

4. Business Rules

- An order's total is the sum of its items (quantity × unit price) plus any service charge percentage.
- Only pending orders can be confirmed; only served orders can be paid.
- Cancelling is allowed only while an order is pending, confirmed, or preparing.
- Data is scoped by branch where applicable (orders, tables, staff, stock).
- A Branch Stock Request and its approval or rejection reserve no stock and make no inventory adjustment; actual balances change only on successful fulfillment.
- Branch Stock Requests are distinct from the older generic Item Requests workflow. Both workflows remain supported and must not be conflated in UI, authorization, reporting, or audit evidence.
- All Branch Stock Request reads and writes must enforce restaurant isolation, plus destination-branch isolation for Branch Storekeepers and Managers.

5. Verification Status

Functional development is active and major workflows have been exercised manually. Complete automated regression evidence for every role, branch boundary, order transition, payment, inventory operation, notification, report, and export is not yet recorded.

Production release remains blocked until the verification matrix in document 12 is complete.